Dilathma Vithana

Computing undergraduate · EEG & brain–computer interface machine learning · Application support engineer

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PROFILE

Final-year BSc (Hons) Computing student, studying part-time while working full-time as an application support engineer at LSEG. My research interest is making motor imagery brain–computer interfaces reliable for every user, especially the new and weaker users that average-accuracy results hide. I combine reproducible machine learning on EEG with production engineering experience, and I aim to pursue graduate research leading to a PhD in neurotechnology.

RESEARCH EXPERIENCE

Cross-Subject Motor Imagery EEG Decoding: Worst-User Benchmark Oct 2026 – Present

Independent research, preliminary study for final-year project · github.com/Dvithana/eeg-signal-toolkit

- Built a reproducible Python pipeline (MNE-Python, pyRiemann, scikit-learn) for left- vs right-hand motor imagery on the PhysioNet EEGMMIDB dataset (**104 subjects**, 4,680 trials, 64 channels).
- Compared CSP+LDA, Riemannian tangent-space classification and per-subject Riemannian re-centering under within-subject and leave-one-subject-out evaluation, with no test-subject data used for training or tuning.
- Re-centering raised new-user balanced accuracy from **62–63% to 70%** (paired Wilcoxon, $p < 0.0001$), yet the lowest 20% of users reached only **54%** and 54% of users stayed below the 70% usability threshold.
- Reported worst-user metrics, paired statistics and limitations (e.g., 45 trials per subject make within-subject estimates unreliable); released code, tests and results openly.

Final-Year Project: Robust Motor Imagery EEG Decoding Under Cross-Subject and Cross-Session Shift 2026 – 2027

BSc (Hons) Computing, Coventry University (NIBM) · proposal submitted

- Proposes re-weighting EEGNet training towards users who score below what their own EEG allows, compared with GroupDRO, VREx and removing weak training users, on OpenBMI (54 people, two sessions each).
- Evaluates worst-20% user accuracy, between-day change and a calibration-budget curve (0, 10, 20 and 40 trials) for new users.

Machine Learning for Coral Reef Bleaching Recovery 2025 – Present

Undergraduate research project, NIBM

- Compared regression models (linear, polynomial, SVR) with leave-one-out cross-validation on Seychelles reef recovery data.
- Audited data provenance against the original authors' repository and found that the predictor columns were model simulations rather than field measurements; documented the issue and proposed re-scoping to the HeatCRD global reef database (29,205 surveys at 12,266 sites).

WORK EXPERIENCE

Application Support Associate Engineer Sep 2026 – Present

LSEG (London Stock Exchange Group), Colombo · full-time

- Support production applications: investigate and resolve incidents, analyse logs and errors, and troubleshoot on Linux and SQL.
- Initiated and built a Python tool that uses text embeddings to match new incidents to similar past incidents by their error messages, surfacing previous root causes and workarounds to speed up diagnosis; currently extending it.

Intern, Application Management May 2026 – Aug 2026

LSEG (London Stock Exchange Group), Colombo

- Trained in application monitoring, incident handling and service management processes; promoted to Associate Engineer at the end of the internship.

Web Developer Intern 2025 – 2026

Jadetimes Media USA

- Developed a React (Vite) news and journals front end and a Node.js/Express back end with SQL database integration.

EDUCATION

BSc (Hons) Computing 2026 – expected 2027

Coventry University, delivered at the National Institute of Business Management (NIBM), Sri Lanka · part-time

Higher National Diploma in Software Engineering 2024 – 2025

NIBM · coursework included machine learning, IoT, mobile and enterprise application development

Diploma in Software Engineering 2023 – 2024

NIBM · preceded by Certificate in Software Engineering (2023) · GCE O/L, Good Shepherd B.M.V. (2023): 7 A, 1 B, 1 C

Certificate in Research Methodology 2025

University of Sri Jayewardenepura, Sri Lanka

TECHNICAL SKILLS

ML & signal processing	Python, NumPy, SciPy, pandas, scikit-learn, MNE-Python, pyRiemann, Matplotlib, text embeddings; cross-validation design, statistical testing
Software	Java, C#, C, PHP, JavaScript/TypeScript, React, Next.js, Node.js/Express, Android (Java)
Data	SQL (MySQL, Oracle), MongoDB, Firebase
Systems & tools	Linux, Bash, Git/GitHub, Nginx, Vercel, pytest, Jupyter, Google Colab

SELECTED ENGINEERING PROJECTS

Special Graphics · Next.js, TypeScript, Nginx
Platform connecting businesses with design experts; multi-page responsive app deployed to Vercel and Nginx.

Automated Response System & App · IoT, C++, Java, Firebase
Sensor system (BMP280, MPU6050, water level) with real-time Firebase storage and an Android dashboard.

Timber Company Management System · PHP, MySQL
Inventory, orders, suppliers, deliveries and finance for a real timber business (diploma final project).

UrbanFood · Java, MongoDB
E-commerce and food delivery management with order processing and sales reporting.

CERTIFICATIONS, MEMBERSHIPS & ACTIVITIES

- Linux Crash Course with Labs – KodeKloud (2025); Linux Unhatched – Cisco (2025)
- MindCraft AI Bootcamp – GitHub Student Chapter (2025)
- Oracle Academy: Java Deep Dive; Oracle Cloud & Databases (2025)
- SQL (Basic) – HackerRank; Basic Oracle SQL – Udemy (2025)
- IEEE Member (Jan 2025 – Present)
- Outreach: led “How to Get Ready for Exams & Road to Future with ICT” for Grade 10 students
- Languages: English, Sinhala

REFERENCE

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